

Apostol, Chapter 1 - Exercise 11

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Prove that $n^4 + 4$ is composite if $n > 1$.

For $n > 1$, $n^4 + 4$ can be decomposed in two factors (both bigger than one) in the following way:

$$\begin{aligned}n^4 + 4 &= (n^2 + 2)^2 - 4n^2 \\&= (n^2 + 2)^2 - (2n)^2 \\&= (n^2 + 2 - 2n)(n^2 + 2 + 2n) \\&> 1 \times 5\end{aligned}$$